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DSC 288R
Week 3 - EDA

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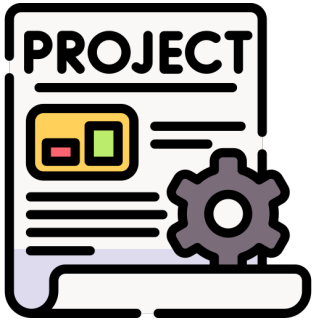
Milestones

Week	Tasks
1	Select Your Teammates
2	Submit Project Abstract
3	Create Exploratory Analysis
4	Start Modeling and Evaluation; Submit Project Progress Report
5	Continue Modeling and Evaluation
6	Continue Modeling and Evaluation
7	Prepare Project Video Presentation
8	Document Your Models
9	Peer Review Project Presentations
10	Submit Final Report



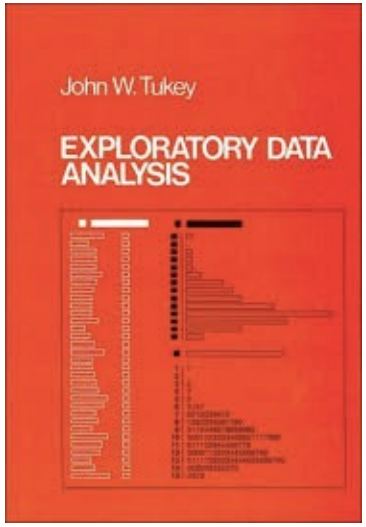
What is EDA?

- A systematic process of understanding data before modeling
- Focuses on structure, quality, distributions, relationships, and anomalies
- Uses statistics, visualization, and domain reasoning

Goal: Gain insights and generate hypotheses, not confirm them

Why do EDA?

- Reveals data quality issues early (degree of missing data, noise, bias)
- Guides feature engineering and model choice
- Prevents invalid assumptions in downstream analysis
- Builds intuition and trust in the dataset



More on What can EDA offer?

- Identifying **outliers** and **missing** values
- Identifying **trends** in time and space
- Uncover **patterns** related to the target (dependent variable)
- Creating **hypotheses** and testing them through experiments
- Enable data-driven decisions
- **Test** underlying assumptions (especially for implementing statistical models)
- Identify **essential and unnecessary features**
- Examine distributions such as examination of Normality
- **Assess data quality**
- **Graphical representation** of data for quick visual analysis

Core Questions EDA Answers

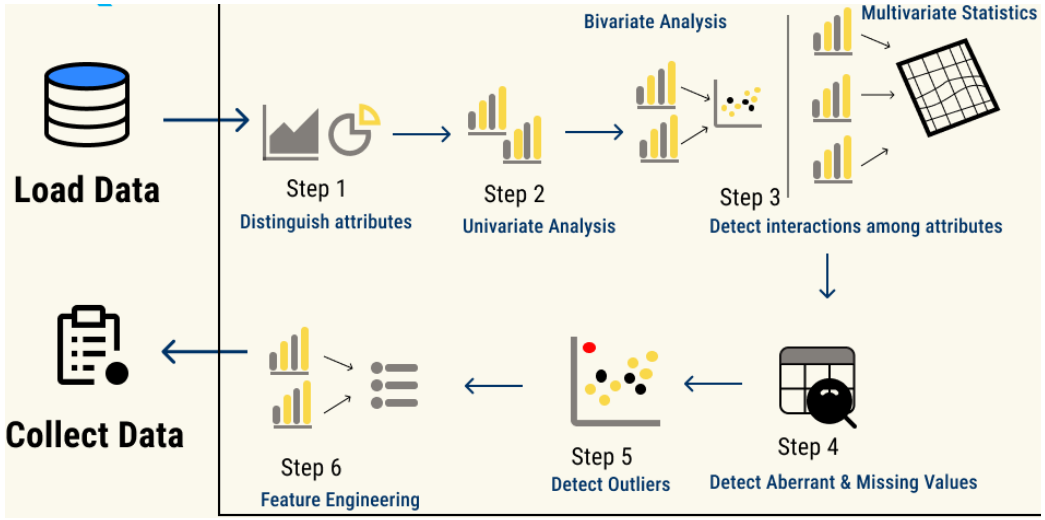
- What data do we actually have?
- How clean, complete, and consistent is it?
- What patterns, trends, or anomalies exist?
- What relationships seem plausible or implausible?

EDA vs Descriptive Analytics

- Descriptive analytics reports known metrics (means, KPIs)
- EDA asks open-ended questions about the data
- EDA is iterative and adaptive, not one-shot
- EDA often combines visuals with narrative reasoning

A Typical EDA Workflow

1. **Understand data sources** – completeness, freshness, quality, meaning etc.
2. The **variables and their distributions** – continuous (distributions) or Categorical (unique values)
3. Describe **Anomalies** – outliers, errors, duplicates



The Data Health Check (Pre-Analysis)

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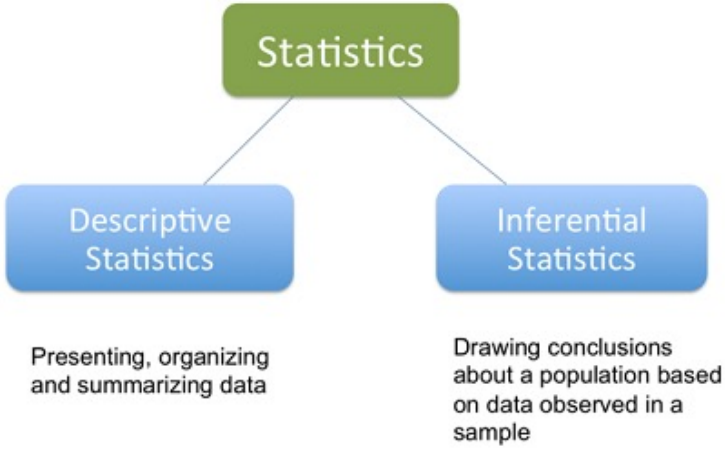
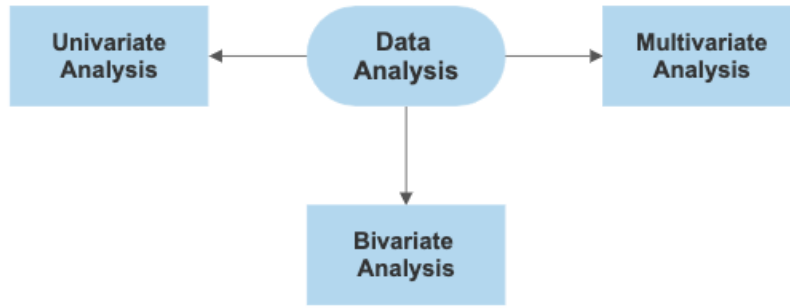
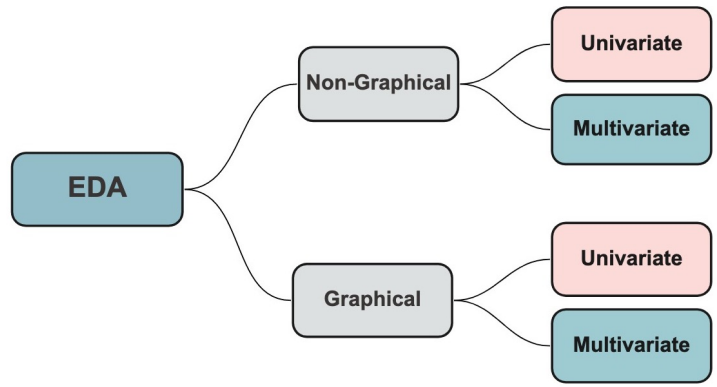
- Before visualization, you must assess data hygiene.
- **Shape:** How many rows (observations) and columns (features)?
- **Types:** Are numerical columns accidentally stored as strings?
- **Missingness:** Heatmaps of null values—is missing data random or systematic?
- **Uniqueness:** Identifying duplicate rows or columns with zero variance (single value).

Types of EDA


Statistics
Descriptive Statistics
Inferential Statistics
Presenting, organizing and summarizing data
Drawing conclusions about a population based on data observed in a sample


Univariate Analysis
Data Analysis
Multivariate Analysis
Bivariate Analysis


EDA
Non-Graphical
Univariate
Multivariate
Graphical
Univariate
Multivariate

Univariate Non-Graphical Analysis

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One variable at a time - (e.g., "What is the average age of our customers?")

Descriptive

- mean, median, mode, minimum and maximum, interquartile range, variance, coefficient of variance, standard deviation, quartiles etc.
- Skewness
- Kurtosis

Inferential

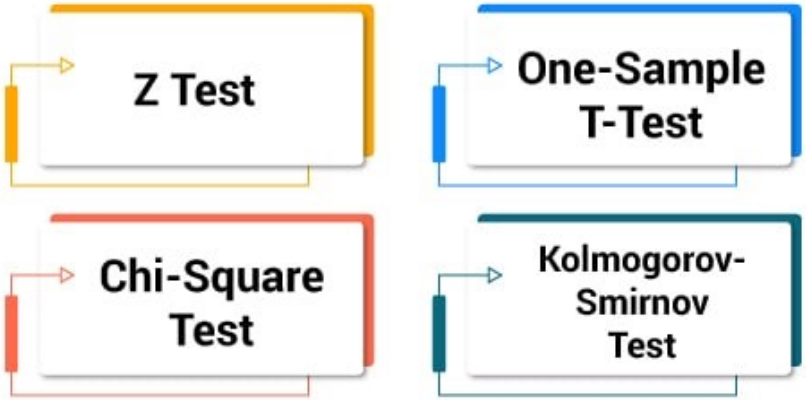
- One sample t-test
- Dependent t-test
- One-way ANOVA (F-test)
- Number of missing values
- Identification of outliers

Univariate Graphical Analysis

- **Histogram** – how are values distributed?
- **Frequency Bar/Pie Chart** – Frequencies of different categories
- **Box Plot** – Helps visualize minimum, maximum, median, and quartiles. It is beneficial in identifying outliers in the data.
- **Density curve** - is the data is bimodal, normally distributed, skewed, etc.?

Inferential Testing

Example: Checking if the price of sugar has statistically significantly risen from the generally accepted price by using sample survey data. Hypothesis tests such as the Z or T-test answer such questions.



Bi/Multi-variate Analysis

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Analyze two or more variables using descriptive and mainly inferential statistics to <u>understand their relationship</u>.

(e.g., "Does age correlate with spending?", "How does age and location combined affect spending?")

<u>Non-Graphical</u>

- Aggregation and summarization using categorical and a numerical variable
- Regression
- Correlation Coefficient
- PCA
- Chi-Square Test

<u>Graphical</u>

- Scatter Plot
- Heat Map
- Stacked / Dodged Bar Chart
- Multiline Chart

EDA Libraries

Python

- Pandas Profiling
- Sweetviz
- Dataprep
- Klib
- Datatile
- SpeedML etc.

R

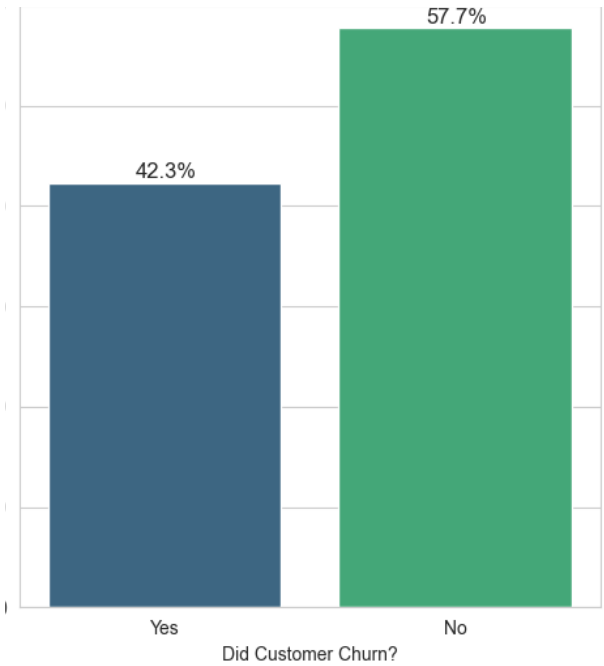
- Data Explorer
- SmartEDA
- DataMaid
- Ggally
- Skimr etc.

Use Case – Telco Customer Churn

Use Case Introduction

- **Scenario:** A telecommunications company wants to know why customers are leaving ("churning").
- **The Dataset:** 7,000 customers, 21 features (Demographics, Services, Account Info).
- **The Target Variable:** Churn (Yes/No).
- **Objective:** Use EDA to find the strongest signals predicting churn *before* building a machine learning model.

Step 1 – Univariate Analysis



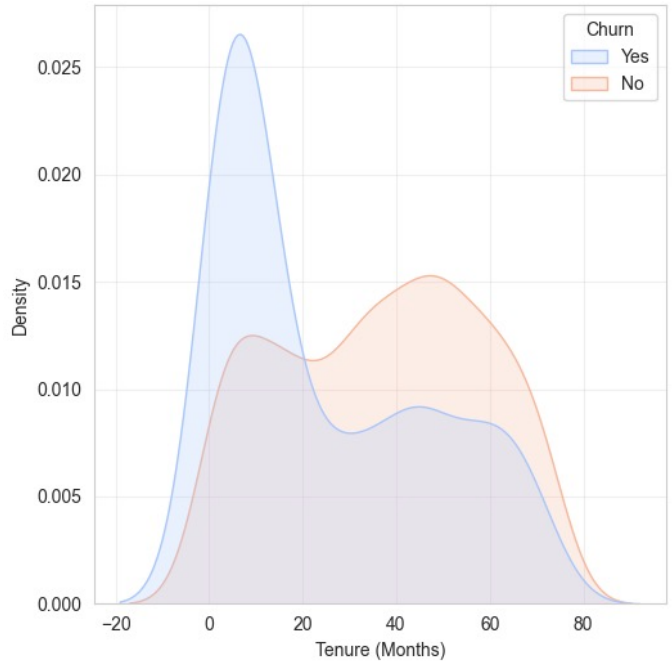
Observation: 58% "No", 22% "Yes".

Insight: The dataset is imbalanced.

Actionable Takeaway: We cannot use "Accuracy" as our primary metric for modeling later; we must use F1-Score or AUC-ROC. We might need resampling techniques (SMOTE).

Step 2: Numerical Analysis of Tenure

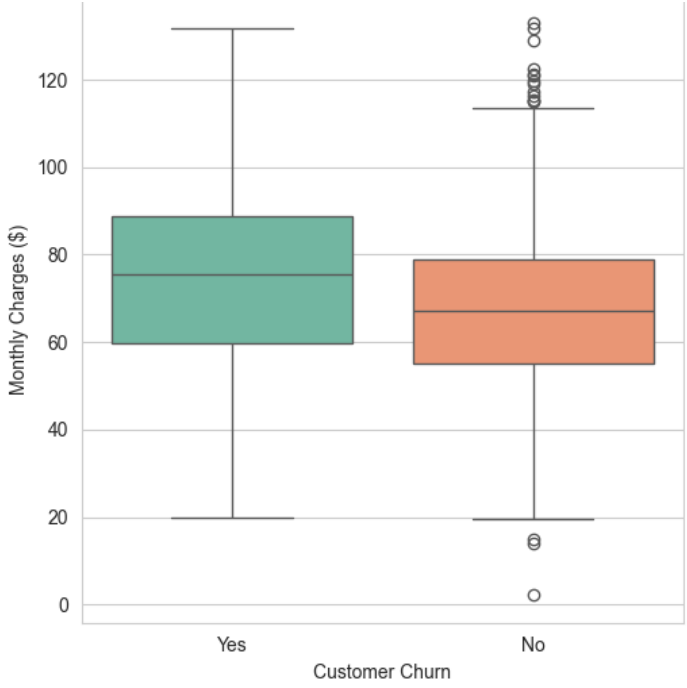
KDE (Kernel Density Estimate) plot of Tenure (months stayed) separated by Churn class.



- **Observation:** The Churn group (Yes) has a massive spike at 1–5 months. The Loyal group (No) has a spike at 70+ months.
- **Insight:** Churn is heavily concentrated in the "onboarding" phase.
- **Actionable Takeaway:** Marketing needs an intervention program specifically for new customers in the first 90 days.

Step 3: Numerical Analysis of Monthly Charges

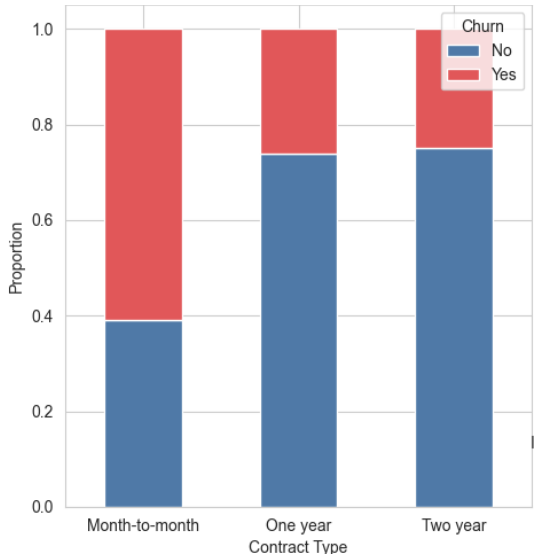
Box plot of MonthlyCharges vs. Churn.



- **Observation:** Churners have a significantly higher median monthly bill ($80) compared to non-churners ($65).
- **Insight:** Price sensitivity is a major driver.
- **Actionable Takeaway:** Investigate if these high charges are due to premium services or legacy pricing plans.

Step 4: Categorical Analysis

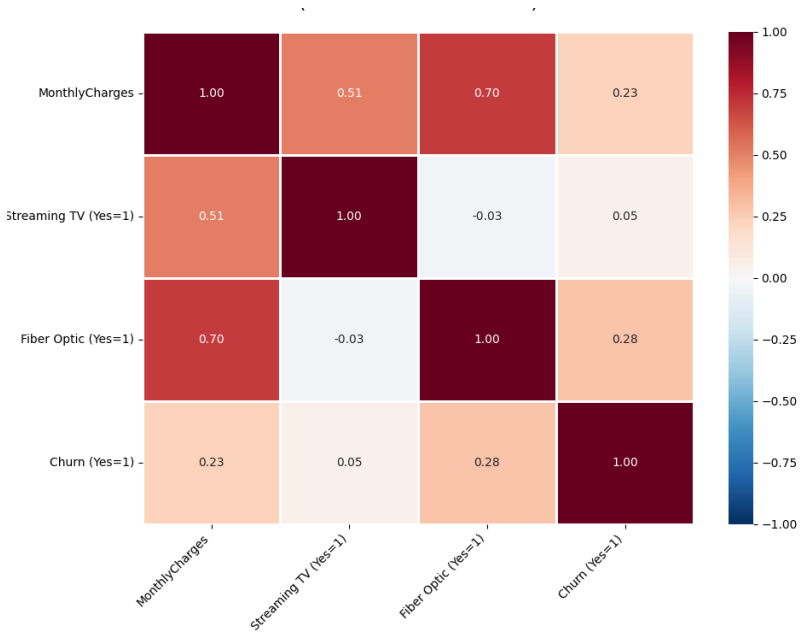
Stacked Bar Chart of Contract type normalized by Churn %.



- **Observation:**
  - Month-to-month contracts: **~60% Churn Rate**.
  - Two-year contracts: **22% Churn Rate**.
- **Insight:** This is the strongest predictor found so far.
- **Actionable Takeaway:** The company should offer incentives (discounts) to migrate month-to-month users to 1-year contracts to lock them in.

Multivariate Analysis

A Heatmap showing correlation between services.



- **High Cost of Fiber:** There is a strong positive correlation between "Fiber Optic" and "Monthly Charges," confirming it is the most expensive service tier.
- **The Churn Driver:** "Fiber Optic" shows the highest positive correlation with Churn (**0.28**), stronger than the correlation between Churn and Monthly Charges alone.
- **Insight:** The problem isn't just "high prices"—it is specifically the **Fiber Optic** product. Customers are paying a premium for Fiber but are churning at the highest rate. This suggests a major "Value for Money" discrepancy or technical instability specific to fiber lines.
- **Actionable Takeaway:** Launch an immediate technical audit of the Fiber network and consider a temporary price-lock promotion for Fiber users to prevent mass exodus.

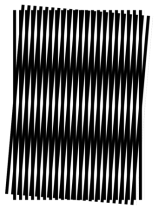
Use Case Summary

Without running a single predictive model, EDA told us:

1. Focus retention efforts on **Month 0–5**.
2. **Fiber Optic** service needs a quality audit.
3. Migrating users to **Yearly Contracts** is the highest ROI strategy.
4. The dataset needs **balancing** before modeling.

When is this effective?

1. **<u>Conveys real insights</u>**: ease of interpretation
2. **<u>Accurate:</u>** Lie factor = size of visual effect/size of data effect.
3. **<u>Efficient</u>**: minimize data-ink ratio and chart-junk
4. **<u>Aesthetics</u>**: must not offend viewer's senses (e.g. moire patterns).



5. **<u>Adaptable</u>**: can adjust to serve multiple needs.

EDA Anti-Patterns – The “average” trap

- **The Mistake:** Relying solely on the mean (average) to describe data.
- **Scenario:** "The average bill is $50."
- **The Reality:** Your data might be bimodal (two humps: lots of $20 bills and lots of $80 bills, but almost no $50 bills).
- **The Fix:** Always visualize the *distribution* (Histogram) along with the mean.

Anti-Pattern 2: Spurious Correlations

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- **The Mistake:** Assuming correlation equals causation.
- **Scenario:** "Ice cream sales correlate with drowning deaths."
- **The Reality:** Both are driven by a third variable: **Summer (Heat)**.
- **The Fix:** Use domain knowledge to sanity-check strong correlations. Don't blindly feature-engineer based on correlation matrices.

Anti-Pattern 3: Anscombe's Quartet

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- **The Concept:** Four datasets that have identical descriptive statistics (mean, variance, correlation) but look completely different when plotted.
- **The Lesson:** If you only look at the summary table, you miss the outliers and non-linear patterns.
- **The Fix: Always visualize.** Summary stats are lossy compression; plots are the raw truth.

Takeaway

EDA is not just making charts; it is the phase where you build your intuition and validate your data quality.

Next Steps: Proceed to Feature Engineering (Handling the outliers and missing values identified during EDA).

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Questions?